

Hard-filtering
germline short
variants

Introduction

Hard Filtering VS VQSR

- **VQSR:** Variant Quality Score Recalibration run on machine learning (Gaussian mixture model (GMM))

Visualizing the distribution of annotation values

Visualization Approach for Hard-Filtering:

- Purpose: Visualize distribution of annotation values in a truth set to derive filtering thresholds.
- Data Origin: Whole genome trio (NA12878, NA12891, NA12892) analyzed using HaplotypeCaller and GenotypeGVCFs.
- Annotation Extraction: VariantsToTable used to extract annotation values for plotting in RStudio.

Parameters:

- QD (Quality by Depth)
- FS (FisherStrand)
- SOR (Strand Odds Ratio)
- MQ (RMSMappingQuality)
- MQRankSum (MappingQualityRankSumTest)
- ReadPosRankSum

Defintions:

QD (Quality by Depth): This is the variant confidence (from the QUAL field) divided by the unfiltered depth of non-hom-ref samples. This annotation is intended to normalize the variant quality in order to avoid inflation caused when there is deep coverage. For filtering purposes it is better to use QD than either QUAL or DP directly.

FS (FisherStrand): This is the Phred-scaled probability that there is strand bias at the site. Strand Bias tells us whether the alternate allele was seen more or less often on the forward or reverse strand than the reference allele. When there little to no strand bias at the site, the FS value will be close to 0.

SOR (Strand Odds Ratio): SOR was created because FS tends to penalize variants that occur at the ends of exons. Reads at the ends of exons tend to only be covered by reads in one direction and FS gives those variants a bad score. SOR will take into account the ratios of reads that cover both alleles.

MQ (RMSMappingQuality): This is the root mean square mapping quality over all the reads at the site. Instead of the average mapping quality of the site, this annotation gives the square root of the average of the squares of the mapping qualities at the site. It is meant to include the standard deviation of the mapping qualities. Including the standard deviation allows us to include the variation in the dataset.

MappingQualityRankSumTest (MQRankSum): It compares the mapping qualities of the reads supporting the reference allele and the alternate allele. A positive value means the mapping qualities of the reads supporting the alternate allele are higher than those supporting the reference allele; a negative value indicates the mapping qualities of the reference allele are higher than those supporting the alternate allele. A value close to zero is best and indicates little difference between the mapping qualities.

ReadPosRankSumTest (ReadPosRankSum): It compares whether the positions of the reference and alternate alleles are different within the reads. Seeing an allele only near the ends of reads is indicative of error, because that is where sequencers tend to make the most errors. A negative value indicates that the alternate allele is found at the ends of reads more often than the reference allele; a positive value indicates that the reference allele is found at the ends of reads more often than the alternate allele. A value close to zero is best because it indicates there is little difference between the positions of the reference and alternate alleles in the reads.

PARAMETERS FOR EACH FILTER:

Parameter	VQSR Threshold	Hard Filtering Threshold	GATK Command Parameter
QualByDepth (QD)	Varies based on model	$\lt 2.0$	$\lt 1.5$
FisherStrand (FS)	$\gt 60$	$\gt 55$	-
StrandOddsRatio (SOR)	$\gt 3$	$\gt 3$	-
RMSMappingQuality (MQ)	$\lt 50$	$\lt 40$	-
MQRankSum	$\lt -2.5$	$\lt -12.5$	-
ReadPosRankSum	$\lt -1.0$ (fail) and $\gt 3.5$ (fail)	$\lt -4$ (fail) and $\gt 4$ (fail)	-
Depth (DP)	-	-	$\lt 10$
Quality (QUAL)	-	-	$\lt 30$

Filtration in our Lab Command

filter-expression "MQ0 >= 5 && ((MQ0 / (1.0 * DP)) > 0.1)":

- MQ0: This represents the count of reads with mapping quality zero.
- DP: This stands for the depth of coverage, which is the total number of reads covering a variant site.

filter-expression "DP < 10":

- DP: This stands for the depth of coverage, which is the total number of reads covering a variant site.

filter-expression "QD < 1.5":

- QD: This stands for "Quality by Depth," which is a normalized variant confidence score adjusted by the depth of coverage (DP).
- $QD = QUAL / DP$

filter-expression "QUAL > 30.0 && QUAL < 50.0":

- This stands for "Quality," which is a Phred-scaled score assigned to each variant by variant calling algorithms. It represents the likelihood that the variant call is accurate.

filter-expression "QUAL < 30.0":

- This stands for "Quality," which is a Phred-scaled score assigned to each variant by variant calling algorithms. It represents the likelihood that the variant call is accurate.